Version 1.1

MISRA Checking User's Guide

Copyright © COSMIC Software 2023, 2024
All Trademarks are the property of their respective owners

MISRA Checking

The **MISRA** checking feature is enabled by compiling with the **+misra** option (or **-pcm**). The parser implements most of the verifications, but some ones must be checked by the linker. The parser then forwards information to the linker through the code generator, the optimizer and the assembler. The information is stored in an extra assembler section named **.misra**, which is ignored by the other tools. The segment name appears in the map file as an empty segment, showing that the **MISRA** checking was enabled for that application.

The **MISRA** rules are divided into directives and rules, and are classified in mandatory, requested and advisory categories. Mandatory and requested rules are reported as **errors**, while advisory rules are reported as **warnings**, unless the compiler is executed with the **+strict** option, all reports becoming errors in such a case.

The compiler allows each directive or rule to be enabled or disabled individually by using a specific **#pragma** directive.

```
#pragma misra enable <list>
#pragma misra disable <list>
```

<list> contains a list of directive/rule number prefixed by the **dir** or **rule** keyword turning the next numbers as directives or rules respectively, or the single keyword **all**. Numbers are considered as rule numbers by default.

The directive:

```
#pragma misra disable 15.3 dir 4.5 4.6 rule 3.1 3.2
```

disables rule 15.3, directive 4.5, directive 4.6, rule 3.1 and rule 3.2,

and the directive:

```
#pragma misra enable all
```

enables all implemented rules.

There is an error message if the directive or the rule number does not exist, but not if the directive or the rule is not implemented.

The directives checked by the linker must be disabled in the source files. The linker informations are forwarded at the end of the compiler process, so the disabled status of these rules is exported at that moment, using then the status of these rules at the end of the file. A linker rule is disabled is at least one file forwards a disabled status.

Another directive is available for the system headers:

```
#pragma misra system
```

This directive disables most checkings and recordings to avoid errors reported from the content of any system header. It operates up to the end of the included file.

The system header files have been adapted with this directive and with a few others changes to be fully accepted by the compiler when the **MISRA** checking is enabled.

Here is the list of all directive and rules from the **MISRA 2012** document. Refer to the **MISRA** document to have a complete description of the directives and rules behavior. The following description is using the following specific status:

Status	Description
Implicit	The situation cannot happen with the Cosmic compiler
User	The check can be done only by the programmer
Not relevant	The feature is not provided by the Cosmic compiler

The other status used are self-explanatory.

Type	Description
Dir 1.1	Not implemented
Dir 2.1	Implicit
Dir 3.1	User
Dir 4.1	User
Dir 4.2	User
Dir 4.3	Implemented in the parser
Dir 4.4	Not Implemented
Dir 4.5	Implemented in the parser
Dir 4.6	Implemented in the parser
Dir 4.7	User
Dir 4.8	Not implemented
Dir 4.9	Not Implemented
Dir 4.10	Not Implemented
Dir 4.11	User
Dir 4.12	User
Dir 4.13	User

Type	Description
Rule 1.1	Not Implemented

Type	Description
Rule 1.2	Implemented in the parser
Rule 1.3	Not Implemented
Rule 2.1	Implemented in the parser
Rule 2.2	Implemented in the parser
Rule 2.3	Implemented in the linker
Rule 2.4	Implemented in the linker
Rule 2.5	Implemented in the linker
Rule 2.6	Implemented in the parser
Rule 2.7	Implemented in the parser
Rule 3.1	Implemented in the parser
Rule 3.2	Implemented in the parser
Rule 4.1	Implemented in the parser
Rule 4.2	Implemented in the parser
Rule 5.1	Implicit
Rule 5.2	Implicit
Rule 5.3	Implemented in the parser
Rule 5.4	Implicit
Rule 5.5	Implemented in the parser
Rule 5.6	Implemented in the linker
Rule 5.7	Implemented in the linker
Rule 5.8	Implemented in the linker
Rule 5.9	Implemented in the linker

 © 2024 COSMIC Software

Type	Description
Rule 6.1	Implicit
Rule 6.2	Implemented in the parser
Rule 7.1	Implemented in the parser
Rule 7.2	Implemented in the parser
Rule 7.3	Implemented in the linker
Rule 7.4	Implemented in the linker
Rule 8.1	Implemented in the parser
Rule 8.2	Implemented in the parser
Rule 8.3	Implemented in the parser
Rule 8.4	Implemented in the parser
Rule 8.5	Implemented in the linker
Rule 8.6	Implicit
Rule 8.7	Implemented in the linker
Rule 8.8	Implemented in the parser
Rule 8.9	Implemented in the linker
Rule 8.10	Implemented in the parser
Rule 8.11	Implemented in the parser
Rule 8.12	Implemented in the parser
Rule 8.13	User
Rule 8.14	Implicit
Rule 9.1	Implemented in the parser
Rule 9.2	Implemented in the parser

Type	Description
Rule 9.3	Implemented in the parser
Rule 9.4	Implemented in the parser
Rule 9.5	Implemented in the linker
Rule 10.1	Implemented in the parser
Rule 10.2	Implemented in the parser
Rule 10.3	Implemented in the parser
Rule 10.4	Implemented in the parser
Rule 10.5	Implemented in the parser
Rule 10.6	Implemented in the parser
Rule 10.7	Implemented in the parser
Rule 10.8	Implemented in the parser
Rule 11.1	Implemented in the parser
Rule 11.2	Implemented in the parser
Rule 11.3	Implemented in the parser
Rule 11.4	Implemented in the parser
Rule 11.5	Implemented in the parser
Rule 11.6	Implemented in the parser
Rule 11.7	Implemented in the parser
Rule 11.8	Implemented in the parser
Rule 11.9	Implemented in the parser
Rule 12.1	Implemented in the parser
Rule 12.2	Implemented in the parser
Rule 12.3	Implemented in the parser
Rule 12.4	Implemented in the parser

 © 2024 COSMIC Software

Type	Description
Rule 13.1	Implemented in the parser
Rule 13.2	Not Implemented
Rule 13.3	Not Implemented
Rule 13.4	Implemented in the parser
Rule 13.5	Implemented in the parser
Rule 13.6	Implemented in the parser
Rule 14.1	Implemented in the parser
Rule 14.2	Implemented in the parser
Rule 14.3	Implemented in the parser
Rule 14.4	Implemented in the parser
Rule 15.1	Implemented in the parser
Rule 15.2	Implemented in the parser
Rule 15.3	Implemented in the parser
Rule 15.4	Implemented in the parser
Rule 15.5	Implemented in the parser
Rule 15.6	Implemented in the parser
Rule 15.7	Implemented in the parser
Rule 16.1	Implemented in the parser
Rule 16.2	Implemented in the parser
Rule 16.3	Implemented in the parser
Rule 16.4	Implemented in the parser
Rule 16.5	Implemented in the parser

Type	Description
Rule 16.6	Implemented in the parser
Rule 16.7	Implemented in the parser
Rule 17.1	Implemented in the parser
Rule 17.2	Implemented in the linker
Rule 17.3	Implemented in the parser
Rule 17.4	Implemented in the parser
Rule 17.5	Implemented in the parser
Rule 17.6	Implicit
Rule 17.7	Implemented in the parser
Rule 17.8	Implemented in the parser
Rule 18.1	Not Implemented
Rule 18.2	Not Implemented
Rule 18.3	Not Implemented
Rule 18.4	Implemented in the parser
Rule 18.5	Implemented in the parser
Rule 18.6	Implemented in the parser
Rule 18.7	Implemented in the parser
Rule 18.8	Implicit
Rule 19.1	Not Implemented
Rule 19.2	Implemented in the parser
Rule 20.1	Not Implemented
Rule 20.2	Implemented in the parser

 © 2024 COSMIC Software

Type	Description
Rule 20.3	Implemented in the parser
Rule 20.4	Implemented in the parser
Rule 20.5	Implemented in the parser
Rule 20.6	Implicit
Rule 20.7	Implemented in the parser
Rule 20.8	Implemented in the parser
Rule 20.9	Implemented in the parser
Rule 20.10	Implemented in the parser
Rule 20.11	Implemented in the parser
Rule 20.12	Not Implemented
Rule 20.13	Implicit
Rule 20.14	Implemented in the parser
Rule 21.1	Not Implemented
Rule 21.2	Not Implemented
Rule 21.3	Implemented in the parser
Rule 21.4	Implemented in the parser
Rule 21.5	Not relevant
Rule 21.6	Implemented in the parser
Rule 21.7	Implemented in the parser
Rule 21.8	Implemented in the parser
Rule 21.9	Not relevant
Rule 21.10	Not relevant
Rule 21.11	Not relevant
Rule 21.12	Not relevant

Type	Description
Rule 22.1	Not Implemented
Rule 22.2	User
Rule 22.3	Not relevant
Rule 22.4	Not relevant
Rule 22.5	Not relevant
Rule 22.6	Not relevant

 © 2024 COSMIC Software